

THIYANAYUGI MARIRAJ

📍 Dortmund, Germany (Open to relocation)

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🌐 LinkedIn

🐙 GitHub

🌐 Portfolio



Summary

AI Engineer and Robotics Specialist architecting the synapse between digital intelligence and physical action. Specializing in Autonomous Robotics, Generative AI, and Computer Vision with expertise in multi-agent systems, LLM frameworks (LangChain, CrewAI, AutoGen), and ROS-based robotic platforms. Proven track record in deploying production-ready AI applications, building collaborative perception systems, and developing end-to-end ML pipelines from research to deployment.

Technical Skills

- **Programming:** Python, C++, MATLAB, Bash, JavaScript, SQL, Rust, Go, TypeScript
- **AI & ML:** PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, NumPy, Pandas, Jupyter
- **Generative AI:** LangChain, HuggingFace, CrewAI, AutoGen, n8n, RAG, Vector DB, Claude Code, REST API
- **Robotics:** ROS/ROS2, Isaac Sim, Gazebo, MoveIt, SLAM, LiDAR, PCL, Coppeliasim, Sensor Fusion
- **Cloud & DevOps:** AWS, Google Cloud, Docker, Kubernetes, GitHub Actions, Grafana, Git, Linux
- **Computer Vision:** YOLOv8, CLIP, Object Detection, 3D Vision, Point Cloud Processing
- **Data Analysis:** Statistical Analysis, Time-Series, Data Visualization, Prometheus
- **Development:** FastAPI, Streamlit, Gradio, Web Development, REST APIs
- **Databases:** SQL, FAISS, ChromaDB, Graph Databases, Vector Search
- **Languages:** English (Fluent), German (B2 in Progress), Tamil (Native)

Education

Technical University Dortmund

Master of Science in Automation and Robotics

- Specialization: AI-driven Systems, Computer Vision, Autonomous Robotics, 6G Collaborative Perception

October 2022 – July 2026

Dortmund, Germany

PSG College of Technology

Bachelor of Engineering in Robotics and Automation

- Core Focus: Industrial Automation, Mobile Robotics, Control Systems, Embedded Systems

July 2018 – May 2022

Coimbatore, India

Professional Experience

Information Processing Laboratory, TU Dortmund

Research Assistant – Machine Learning & Signal Processing

- Designed deep learning models for intelligent signal interpretation in complex electromagnetic systems achieving high accuracy in industrial applications.
- Applied reinforcement learning for adaptive optimization and real-time parameter tuning in dynamic environments.
- Improved signal processing precision in high-frequency industrial applications via supervised ML techniques.

June 2024 – May 2025

Dortmund, Germany

Chair of Material Handling and Warehousing, TU Dortmund

Master's Thesis – 6G Collaborative Robotic Perception

- Developed 6G-enabled collaborative perception for multi-robot systems using mmWave radar technology.
- Implemented Graph Neural Network architectures for real-time occupancy prediction in warehouse environments.
- Conducted comprehensive validation using dual robotic platforms with motion capture ground truth systems.
- Technologies: Python, GNN, mmWave Radar, ROS

January 2025 – July 2025

Dortmund, Germany

Pricol Limited

Robotics Engineering Intern – Autonomous Systems

- Designed and deployed Autonomous Mobile Robot for industrial logistics and inspection automation applications.
- Integrated LiDAR, IMU, and camera systems for robust SLAM-based navigation in dynamic industrial environments.
- Developed real-time path planning and obstacle avoidance algorithms using ROS framework.

November 2021 – June 2022

Coimbatore, India

Featured Projects

Autolearn AI Platform - Production Web Application

- Engineered production-ready web platform with Claude API-powered intelligent chatbot and AI-driven personalized email generation system for course enrollment.
- Deployed on Google Cloud Run with scalable architecture handling real-time user interactions.
- Technologies: JavaScript, Claude API, Google Cloud Run, REST APIs

AI Document Explainer - International Student Tool

- Built production-ready AI application helping international students understand complex official documents using GPT-4 natural language processing.
- Implemented Streamlit-based interactive interface with document upload, processing, and explanation features.
- Technologies: Python, GPT-4, Streamlit, Cloud Deployment

RoboVision-3D: Computer Vision for Indoor Robotics

- Developed comprehensive computer vision system integrating RGB-D cameras and LiDAR for 3D environment mapping and object detection.
- Implemented YOLOv8-based object detection with PCA-driven 3D localization processing over 19M point cloud data points.
- Engineered modular ROS-based architecture for real-time multi-sensor fusion and SLAM-based navigation.
- Technologies: Python, YOLOv8, ROS, OpenCV, Point Cloud Library

AutoGen Content Creation Pipeline - Multi-Agent AI

- Architected enterprise-grade multi-agent AI system automating end-to-end content creation workflows with specialized agents for research, writing, and editing.
- Implemented autonomous agent orchestration using Microsoft AutoGen framework with DeepSeek integration.
- Technologies: AutoGen, DeepSeek, Python, Multi-Agent Systems

Market Intelligence System - CrewAI Multi-Agent

- Developed CrewAI-powered multi-agent system with four specialized agents (Researcher, Analyst, Writer, Reviewer) for autonomous market intelligence gathering and analysis.
- Integrated Claude API and RAG techniques for enhanced reasoning and knowledge retrieval.
- Technologies: CrewAI, Claude, RAG, Python

GraphRAG - Knowledge Graph-Enhanced Retrieval System

- Architected production-ready hybrid retrieval system combining knowledge graphs with FAISS vector search for multi-hop reasoning and complex query resolution.
- Implemented automated entity extraction pipeline using spaCy NER with relationship mapping across documents.
- Achieved sub-second query response times processing 140+ entities with semantic relationship graphs.
- Technologies: Graph Databases, FAISS, Vector Search, Gradio, spaCy

Multi-Modal AI Framework for Robotic Task Automation

- Engineered multi-agent system integrating CLIP vision models, GPT-3.5 NLP, and RAG for intelligent robotic task planning and execution.
- Implemented cross-modal understanding enabling robots to process visual and textual instructions simultaneously.
- Technologies: LangChain, CLIP, GPT-3.5, RAG, Computer Vision

Edge Detection 3D Vision System

- Built 3D vision system with real-time image processing and modular edge detection algorithms using ROS framework.
- Integrated OpenCV and RViz for visualization and processing pipeline optimization.
- Technologies: ROS, OpenCV, RViz, 3D Vision

Certifications

- **AWS Training:** Generative AI Learning Plan for Developers, Amazon SageMaker JumpStart Foundations, No-code Machine Learning and Generative AI on AWS, Building Language Models on AWS
- **Hugging Face:** AI Agents with HuggingFace - Fundamentals of AI Agents
- **DeepLearning.AI:** ChatGPT Prompt Engineering for Developers
- **LinkedIn Learning:** What Is Generative AI?, Generative AI: The Evolution of Thoughtful Online Search

Publications

- Mariraj, T., et al. (2021). **RFID based Human Following Load Carrier**. NVEO - Natural Volatiles & Essential Oils Journal, 8(5). [Link]
- Mariraj, T., et al. (2021). **Development Of Robotic Arm On An Omnidirectional Base**. NVEO - Natural Volatiles & Essential Oils Journal, 8(4). [Link]